PAPER_699: Fornax Ultra-Deep Field: 10,000+ Galaxy Statistics and UQFF

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Abstract

The Fornax constellation hosts one of the deepest Hubble observations, revealing 10,000+ galaxies across $z = 0.1$ to 6.5. Statistical tools including the Schechter luminosity function and UQFF galaxy evolution corrections are derived.

1. Survey Parameters

Parameter	Value
N_{gal}	$\sim 10,000$
Area	11 arcmin²
z range	0.1–6.5
$\bar{z}$	1.8

2. UQFF Galaxy Number Evolution

$$N_{UQFF}(z) = N_{obs}(1+z)^{-1.5}\left(1+\frac{\rho_{SCm}}{\rho_{UA}}\right)(1+f_{TRZ})$$

3. Comoving Volume Element

$$\frac{dV}{dz} = \frac{c}{H(z)}D_c^2(z)\,d\Omega$$

4. Schechter Luminosity Function

$$\phi(L) = \frac{\phi^*}{L^*}\left(\frac{L}{L^*}\right)^{\alpha} e^{-L/L^*}$$

With $\alpha \approx -1.3$ for field galaxies.

5. Hubble Parameter

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3+\Omega_\Lambda}$$

References

- Williams et al. (1996), AJ, 112, 1335 (HDF-S)
- UQFF Framework, Star-Magic Session 174

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